

High-Resolution OLED Microdisplay for Near-Eye Display Applications

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Abstract

High-resolution near-eye displays are essential for next-generation extended reality (XR) systems. Organic light-emitting diode on silicon (OLEDoS) offers high contrast and scalability to ultra-high pixel densities. In this work, we present a high-resolution OLEDoS platform and demonstrate its application in a super multi-view (SMV) near-eye display for mitigating the vergence-accommodation conflict (VAC). An SMV prototype is implemented using a high-resolution OLEDoS panel, and experimental results confirm depth-dependent image formation and accommodation cues. These results highlight the potential of OLEDoS-based SMV displays for improving visual comfort in XR.

Author Keywords

OLEDoS; VAC (vergence-accommodation conflict); super multi-view display

1. Introduction

Extended reality (XR) technologies—including augmented reality (AR), virtual reality (VR), and mixed reality (MR)—are rapidly evolving toward spatial computing platforms that require immersive and visually comfortable near-eye displays. Unlike conventional flat-panel displays, near-eye displays must provide a wide field of view (FoV) and high perceptual image quality within a severely constrained panel size and compact optical form factor. To approach the human visual acuity limit of approximately 60 pixels per degree (PPD), theoretical monocular resolutions exceeding $7K \times 7K$ are required [1,2]. In addition, substantial optical losses introduced by compact near-eye optical systems necessitate extremely high display brightness.

Among various microdisplay technologies—such as LCoS, LCD, OLEDoS, and Micro-LED—organic light-emitting diode on silicon (OLEDoS) has emerged as one of the most promising candidates for near-eye applications. Its self-emissive nature enables inherently high contrast ratios and fast temporal response, while silicon CMOS backplanes support dense pixel integration and sophisticated driving circuitry. Owing to these advantages, OLEDoS has already been adopted in recent high-end XR devices, including Apple Vision Pro and Samsung Galaxy XR.

Most commercial XR systems primarily rely on binocular disparity to provide vergence cues for depth perception. However, for the human visual system to perceive depth naturally, vergence (the inward or outward rotation of the eyes) must be consistent with accommodation (the adjustment of the eye's lens focus). In conventional XR displays, the virtual image plane is fixed, and thus the accommodation response of the eye does not match the vergence demand. This mismatch results in the well-known vergence-accommodation conflict (VAC) [3,4], which is widely recognized as a major source of visual discomfort, eye fatigue,

and dizziness in XR environments.

To overcome this limitation, display architectures that provide enhanced focus cues have been actively investigated [5-8]. We have also been actively exploring multifocal display and super multi-view (SMV) display architectures as potential solutions to the VAC, as they offer complementary advantages and system trade-offs. In particular, SMV displays have attracted significant attention due to its ability to generate multiple viewpoints within the pupil and induce a more natural accommodation response [6-8]. By reconstructing appropriate light field information, SMV systems enable continuous depth perception without relying solely on binocular disparity.

In this presentation, we introduce our ongoing development of high-resolution OLEDoS platforms and demonstrate their applicability to SMV near-eye displays. In particular, we present a benchtop SMV prototype based on a high-resolution OLEDoS panel combined with a micro-lens array (MLA), along with an elemental image array (EIA) generation method based on backward ray tracing. Through experimental validation, we demonstrate successful depth-dependent image formation and accommodation cues, highlighting the potential of OLEDoS-based SMV displays as an effective approach for mitigating VAC in next-generation XR systems.

2. OLED microdisplay

We have continuously developed high-resolution OLEDoS platforms based on silicon CMOS backplanes combined with white OLEDs and color filters [9-11]. Through systematic optimization of the OLED device structure and advanced color filter process technologies, both brightness and color gamut have been significantly improved to meet the stringent requirements of near-eye display applications.

Our latest OLEDoS, shown in Figure 1, features a 1.03-inch panel with a resolution of 2560×2560 . The display employs a hexagonal RGB subpixel structure arranged in a delta configuration, with a pixel pitch of $7.2 \mu\text{m}$, enabling high pixel density suitable for near-eye applications. The panel is fabricated based on the RDP1000X CMOS backplane developed by RAONTECH and adopts a white OLED combined with color filters. The measured color gamut reaches 96.8% of the DCI-P3 standard, demonstrating its capability to deliver vivid and accurate colors required for immersive XR experiences.

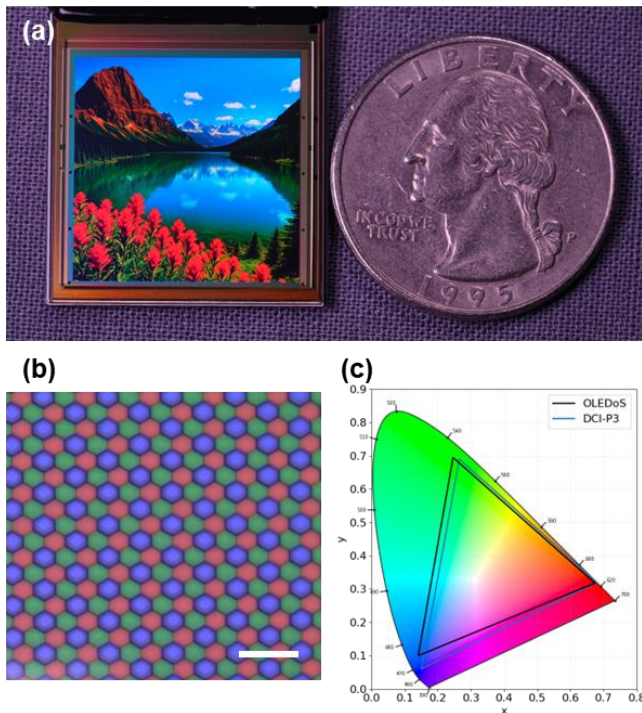


Figure 1. (a) A 1.03-inch OLEDoS panel with a resolution of 2560×2560 used for the SMV display. (b) Optical microscope image of a hexagonal sub-pixel structure with a delta RGB arrangement. Scale bar, 10 μm . (c) Measured color gamut of the OLEDoS compared with the DCI-P3 standard.

3. OLEDoS-based SMV near-eye display

3.1. Benchtop prototype of the SMV near-eye display system

To effectively resolve the VAC, it is essential to provide correct or adaptive accommodation cues to the human visual system. Various approaches have been proposed to address this issue, among which SMV displays have emerged as a promising solution due to their ability to induce a natural accommodation response by forming multiple viewpoints within the pupil. In this work, we focus on an SMV-based approach and implement it using a spatial multiplexing scheme that combines a high-resolution OLEDoS panel with an MLA.

An SMV near-eye display prototype was constructed using the 1.03-inch OLEDoS panel described above, integrated with an MLA (#630, Fresnel Technologies), as shown in Figure 2. The system adopts a spatial multiplexing configuration in which the high pixel density of the OLEDoS panel is utilized to generate multiple directional light rays through the MLA, enabling the formation of multiple viewpoints.

The benchtop prototype also includes a custom-designed driving board for the OLEDoS panel. In addition, a smartphone camera was incorporated into the setup to capture and evaluate the reconstructed images. To achieve high-quality SMV image formation, precise optical alignment between the OLEDoS panel and the MLA is critical. Therefore, multiple precision translation stages were employed to enable fine adjustment of their relative

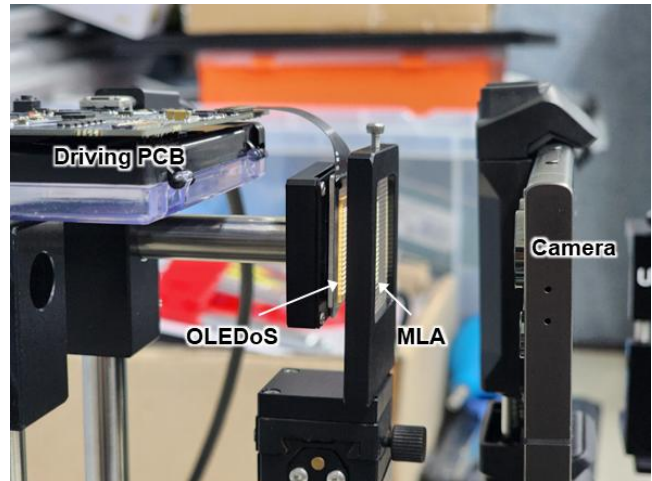


Figure 2. Benchtop prototype of the SMV near-eye display system integrating the OLEDoS panel and the MLA.

position and separation distance, ensuring accurate light field reconstruction.

3.2. Elemental image array

EIAs were generated using a backward ray tracing approach that accounts for both the pixel pitch of the OLEDoS panel and the optical parameters of the MLA. In this method, a single EIA is synthesized by reconstructing the ray distribution (i.e., light field) passing through the MLA in an SMV configuration. The input parameters include the panel resolution and pixel pitch, the MLA lenslet pitch and focal length, and the eye-MLA distance (eye relief), all defined within a virtual 3D optical configuration of the display panel and MLA. The resulting SMV display forms multiple viewpoints within the user's pupil, allowing light rays from different views to overlap when the eye focuses at the intended virtual object distance. Using this approach, a near-eye display capable of covering a depth range from approximately 20 cm to optical infinity was achieved.

For each panel pixel, the center position of the corresponding lenslet is determined based on the pixel location, and a target point on the reference depth plane is defined. This mapping establishes the ray direction that the pixel is intended to reproduce after passing through the lenslet. A ray is then traced from the MLA plane into the virtual scene along the defined direction, and the color value is assigned by sampling the material or texture at the first intersection point. This sampled value is then mapped to the corresponding pixel in the EIA.

Given a pixel pitch of $7.2 \mu\text{m}$ for the OLEDoS and a lenslet pitch of 1 mm for the MLA, approximately 139 pixels correspond to a single lenslet. In addition, since the active area of the OLEDoS panel is $18.7 \text{ mm} \times 18.7 \text{ mm}$, the resulting EIA consists of an array of 18×18 elemental images, each with a size of $1 \text{ mm} \times 1 \text{ mm}$.

Figure 3 illustrates two types of EIAs used in this work. The first type corresponds to a simple text-based scene, in which two words are positioned at different depths, resulting in the representation of two discrete focal planes. In contrast, the second type represents a scene containing volumetric 3D objects placed at

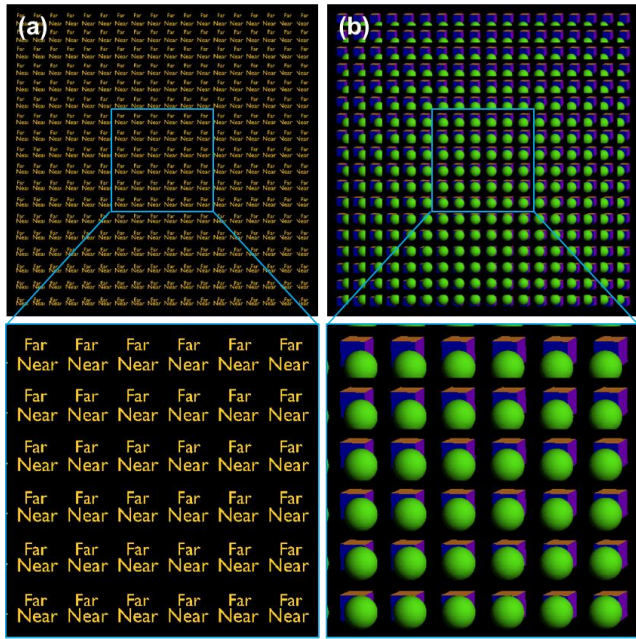


Figure 3. EIAs for the SMV display. (a) EIA for a two-depth image comprising the words “Near” and “Far.” (b) EIA for a scene containing volumetric objects.

different depths. This configuration enables the generation of continuous depth variations, rather than discrete focal planes, thereby demonstrating the capability of SMV to reproduce more realistic three-dimensional scenes.

3.3. Experimental verification

The generated EIAs were displayed on the OLEDoS panel and optically reconstructed through the MLA. The resulting images were captured using a camera, as shown in Figure 4. For the two-depth text EIA, depth-dependent focus cues are clearly observed. When the camera is focused at a near distance (0.25 m), the word “Near” appears sharp, while the word “Far” is blurred due to defocus. Conversely, when the focus is adjusted to a farther distance (2 m), “Far” becomes sharp and “Near” appears blurred. This behavior confirms that correct accommodation cues are successfully reproduced by the SMV display.

Similar results are obtained for the scene containing volumetric objects. When the camera is focused at a near distance (0.2 m), the front sphere is clearly in focus, whereas the rear cube appears blurred. As the focus is shifted to a farther distance (5 m), the cube becomes sharp while the sphere is defocused. These results demonstrate continuous depth-dependent image formation and natural defocus blur, indicating the capability of the proposed system to provide realistic three-dimensional visual cues.

Based on these results, we are currently extending the system to a binocular configuration to simultaneously provide both vergence and accommodation cues for more complete depth perception. In addition, ongoing work includes the development of a system capable of real-time video rendering, moving beyond static image demonstrations.

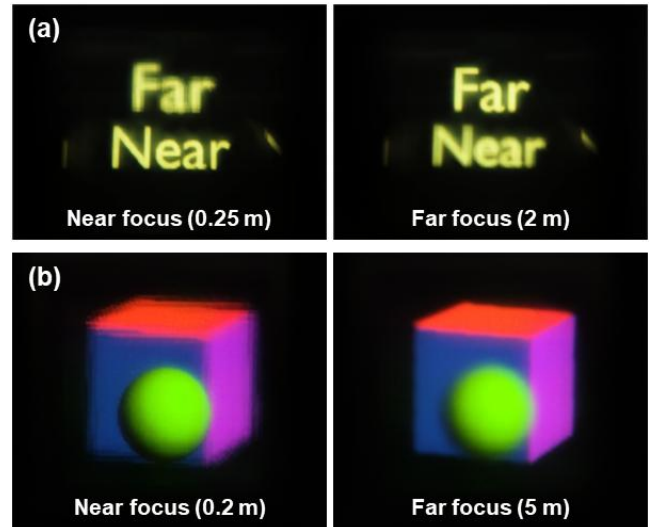


Figure 4. Camera-captured experimental results demonstrating depth-dependent defocus blur. (a) Two-depth image captured with the camera focused at 0.25 m (left) and 2 m (right). (b) Scene containing a sphere and a cube captured with the camera focused at 0.2 m (left) and 5 m (right).

4. Conclusion

The results presented in this work demonstrate that high-resolution OLEDoS provides an effective platform for SMV near-eye display systems aimed at improving visual comfort. By combining a high pixel density OLEDoS panel with an MLA and a backward ray tracing-based EIA generation method, accurate accommodation cues were successfully reproduced, enabling depth-dependent image formation and natural defocus blur. These results confirm the potential of SMV displays to mitigate the VAC, which remains a key limitation in current XR devices.

Beyond the specific prototype demonstrated in this study, the proposed approach highlights the scalability and flexibility of OLEDoS for advanced light field-based display architectures. With its capability to support ultra-high resolution, high brightness, and compact integration, OLEDoS is well positioned to serve as a core display technology for next-generation XR and spatial computing systems requiring both high image quality and enhanced visual comfort.

5. Acknowledgements

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